

```
C BT_array.c C 2a.c X C 2d.c C 2e.c C 2f.c C 2g.c C 2c.c C 2b.c C BT_linkedli
SEMESTER 3 > C 2a.c > main()
1 #include <stdio.h>
2 #include <string.h>
3 int main(){
4     char name[10];
5     strcpy(name, "Amrit");
6     printf("%s | Buffer size: %zu bytes\n", name, sizeof(name));
7     return 0;
8 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\" ; if ($?) { gcc 2a.c -o 2a } ; if ($?) { .\2a }
Amrit | Buffer size: 10 bytes
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3>
```

```
C BT_array.c C 2a.c C 2d.c C 2e.c C 2f.c C 2g.c C 2c.c C 2b.c X C BT_linkedli
SEMESTER 3 > C 2b.c > main()
1 #include <stdio.h>
2 #include <string.h>
3 int main(){
4     char name[10];
5     strcpy(name, "ABCDEFGHJKLMNPO");
6     printf("Value: %s\n", name);
7     return 0;
8 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\" ; if ($?) { gcc 2b.c -o 2b } ; if ($?) { .\2b }
Value: ABCDEFGHJKLMNPO
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3>
```

```
C BT_array.c C 2a.c C 2d.c C 2e.c C 2f.c C 2g.c C 2c.c X C 2b.c C BT_linkedli
SEMESTER 3 > C 2c.c > main()
1 #include <stdio.h>
2 #include <string.h>
3 struct StudentData {
4     char buffer [8];
5     char status [8];
6 };
7 int main(){
8     struct StudentData data;
9     strcpy(data.status, "SAFE");
10    printf("BEFORE: Status = %s\n", data.status);
11    strcpy(data.buffer, "AAAAAAAAHACKED"); // oversized copy into buffer
12    printf("AFTER: Buffer = %s\n", data.buffer);
13    printf("AFTER: Status = %s\n", data.status);
14    return 0;
15 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\" ; if ($?) { gcc 2c.c -o 2c } ; if ($?) { .\2c }
BEFORE: Status = SAFE
AFTER: Buffer = AAAAAAAAAHACKED
AFTER: Status = HACKED
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3>
```

```
C BT_array.c C 2a.c C 2d.c X C 2e.c C 2f.c C 2g.c C 2c.c C 2b.c C BT_linkedli
SEMESTER 3 > C 2d.c > main()
1 #include <stdio.h>
2 int main(){
3     char username[10];
4     printf("Enter username: ");
5     scanf("%9s", username); // Restricted to 9 chars + '\0'
6     printf("Username: %s\n", username);
7     return 0;
8 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\" ; if ($?) { gcc 2d.c -o 2d } ; if ($?) { .\2d }
Enter username: Amrit
Username: Amrit
PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3>
```

```
C BT_array.c C 2a.c C 2d.c C 2e.c X C 2f.c C 2g.c C 2c.c C 2b.c C BT_linkedli
SEMESTER 3 > C 2e.c > main()
1 #include <stdio.h>
2 int main()
3 {
4     char username[10];
5     printf("Enter username: ");
6     fgets(username, sizeof(username), stdin);
7     printf("Username: %s", username);
8     return 0;
9 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\" ; if (\$?) { gcc 2e.c -o 2e } ; if (\$?) { .\2e }

Enter username: Amrit Mahajan
Username: Amrit Mah

PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3>

```
C BT_array.c C 2a.c C 2d.c C 2e.c C 2f.c X C 2g.c C 2c.c C 2b.c C BT_linkedli
SEMESTER 3 > C 2f.c > main()
1 #include <stdio.h>
2 #include <string.h>
3 struct StudentData {
4     char buffer [8];
5     char status [8];
6 };
7 int main()
8 {
9     struct StudentData data;
10    strcpy(data.status, "SAFE");
11    printf("Address of buffer: %p\n", (void *)data.buffer);
12    printf("Address of status: %p\n", (void *)data.status);
13    strcpy(data.buffer, "AAAAAAAHACKED");
14    printf("After overflow, Status = %s\n", data.status);
15    return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\" ; if (\$?) { gcc 2f.c -o 2f } ; if (\$?) { .\2f }

Address of buffer: 0061FF10
Address of status: 0061FF18
After overflow, Status = HACKED

PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3>

```
C BT_array.c C 2a.c C 2d.c C 2e.c C 2f.c C 2g.c X C 2c.c C 2b.c C BT_linkedli
SEMESTER 3 > C 2g.c > main()
1 #include <stdio.h>
2 #include <string.h>
3 int main()
4 {
5     char buffer[10];
6     snprintf(buffer, sizeof(buffer), "%s", "ABCDEFGHJKLMNOPQRSTUVWXYZ");
7     printf("Stored value: %s\n", buffer);
8     return 0;
9 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\MasterJi\Desktop\DSA Programs> cd "c:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3\" ; if (\$?) { gcc 2g.c -o 2g } ; if (\$?) { .\2g }

Stored value: ABCDEFGHI

PS C:\Users\MasterJi\Desktop\DSA Programs\SEMESTER 3>